ABES Engineering College, Ghaziabad

Department of ASH

Estd. 2000

Lab Manual

Year/ Sem. : 1^{st} year/ I sem.

Subject Name : Web Designing

Subject Code : 25VA151

Faculty Name : Mr. Jitendra Kumar Chauhan,

Approved by AICTE, New Delhi & Affiliated to Dr. APJ Abdul Kalam Technical University, UP, Lucknow

NBA Accredited Courses (CSE, IT, ECE, EN)

19^{th} KM Stone, NH-24, Ghaziabad-201009, UP

ABES Engineering College, Ghaziabad

Estd. 2000

VISION

To take ABES Engineering College to such a level that, it is at par with the leading institutions of the world in providing leadership to the international education system and be amongst the top-rated institutions of the world by providing a transformative education to create leaders and innovators embedded in traditional Indian values.

MISSION

- To create an ambiance for healthy teaching learning process.
- To nurture the students and infuse in them:
  - A passion to excel professionally
  - A spirit to be of utmost use to the industry, corporate sector and the society at large
  - An intense desire to take challenging responsibilities and leadership roles
  - A craving to be wholesome good human beings
- To develop an environment for creating new knowledge through research and by thriving to explore innovative ideas.

QUALITY POLICY

To continuously thrive to provide a congenial and wholesome academic environment and a healthy culture for faculty, staff and students which would motivate teacher's full participation with passion and develop an intense desire in the students to acquire comprehensive education and hence become a useful and confident human resource for the industry and academia.

VALUES

Respect – We treat people the way we want to be treated

Integrity – We believe in doing things right when no one is watching

Accountability – We take full responsibility of our actions and are accountable for them

Transparency –We communicate openly amongst ourselves and all our stakeholders to avoid any misconception.

Excellence – We endeavor to think out-of-the-box and deliver maximum value to all our Stakeholders.

ABES Engineering College, Ghaziabad

Department of ASH

VISION

The department of ASH has a distinct vision to nurture and inculcate knowledge and understanding of engineering sciences amongst the budding technocrats and groom the students with intellectual and professional strength. ASH is an anvil to chisel future professionals with the skills of engineering, imbibed with social, ethical and moral values, so that the pupils can expedite their efforts to build technologies from knowledge and be a part of the workforce, instrumental in the development of the nation's technological and economic progress.

MISSION

M1: To play a pivotal role in laying the foundations of engineering education by building the professional strengths of the students in basic sciences and engineering disciplines and strengthening the foundation of aspiring engineers.

M2: To provide a unique learning environment for the students, preparing the students at par with the global technical workforce, building confidence, character and human values.

M3: To upgrade the quality of technical education and produce dynamic engineers and entrepreneurs for the country, with a sole aim to spot ABESEC amongst the best educational institutions of global standards

ABES Engineering College, Ghaziabad

Department of ASH

GENERAL RULES & INSTRUCTIONS OF LAB

1. Students are advised to come to the laboratory at least 5 minutes before (to the starting time), those who come after 5minutes will not be allowed into the lab.
2. Plan your task properly much before to the commencement, come prepared to the lab with the synopsis / program / experiment details.
3. Student should enter into the laboratory with:

    a) Laboratory observation notes with all the details (Problem statement, Aim, Algorithm, Procedure, Program, Expected Output, etc.,) filled in for the lab session.

    b) Laboratory Record updated up to the last session experiments and other utensils (if any) needed in the lab.

    c) Proper Dress code and Identity card.

Do's

1. Switch off mobile phone in the lab.
2. Bring your lab record with you.
3. Always come to the lab in time.
4. Maintain silence and be attentive.
5. Keep individual records of observations.
6. Always turn off the computer before leaving the lab.
7. Return all the material issued to you before leaving the lab.
8. Read the instructions manual carefully before performing any experiment.

Don'ts

1. Eat or drink inside the lab.
2. Overcrowd near any equipment.
3. Make any mark on instruction manual.
4. Drag the chair/stool inside the lab.
5. Leave the lab till the class has concluded.
6. Take any material outside the lab with you.
7. Touch any machine/equipment/model without supervision.

Index:

Sr. No	Title of Experiment	KL	CO	Software Used	Page No
1	Create a Web Page for Course Curriculum: Use of Headings, Paragraphs, Bold, Italic, Underline, and Special Characters, Ordered List, Unordered List, Description List, Nested List, and Mixed Lists.	**K2**	**CO1**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**7-10**
2	Create a Web Page for a News Website with Embedded YouTube Video, Hyperlinks, and Navigation Structure, and Insert Multimedia in a Webpage.	**K2, K3**	**CO1**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**11-15**
3	Create Web page for College timetable: Use of tables with rowspan, colspan, headers, captions, and nested tables	**K3**	**CO2**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**16-18**
4	Create Web page for Hotel room booking form: Use of input fields, labels, select, textarea, and buttons.	**K3**	**CO2**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**19-21**
5	Create Web page for Student Registration Form with Email Validation: Apply form attributes (method, action, placeholder, required)	**K3**	**CO2**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**22-24**
6	Create Web page for E-commerce Product Page: Use of CSS for style text, colors, fonts, margin, border, padding, width, and height	**K3**	**CO3**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**25-28**

7	Create Web page for Navigation Menu: .....Use of block, inline, inline-block, float, and clear	**K3, K4**	**CO4**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**29-31**
8	Create Web page for Student Profile Card: design layouts using flexbox (justify-content, align-items, flex-direction)	**K4**	**CO4**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**32-34**
9	Create Web page for Responsive Product Grid: create mobile-friendly designs with breakpoints.	**K4**	**CO4**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**35-37**
10	Create Web page for E-commerce Style Navbar: create a responsive navbar with CSS and media queries	**K4**	**CO4**	• Text Editor (VS Code, Sublime Text, Notepad++) • Web Browser (Chrome, Edge, Firefox)	**38-41**

Experiment-1

Create a Web Page for Course Curriculum: Use of Headings, Paragraphs, Bold, Italic, Underline, and Special Characters, Ordered List, Unordered List, Description List, Nested List, and Mixed Lists.

Objectives

1. To design a simple course curriculum webpage using HTML.
2. To practice the use of basic text formatting tags (bold, italic, underline, special characters).
3. To learn the implementation of ordered, unordered, description, nested, and mixed lists.
4. To understand webpage content structuring using headings and paragraphs.

Tools Required

- Computer / Laptop with any OS
- Text Editor (VS Code, Sublime Text, Notepad++)
- Web Browser (Chrome, Edge, Firefox)

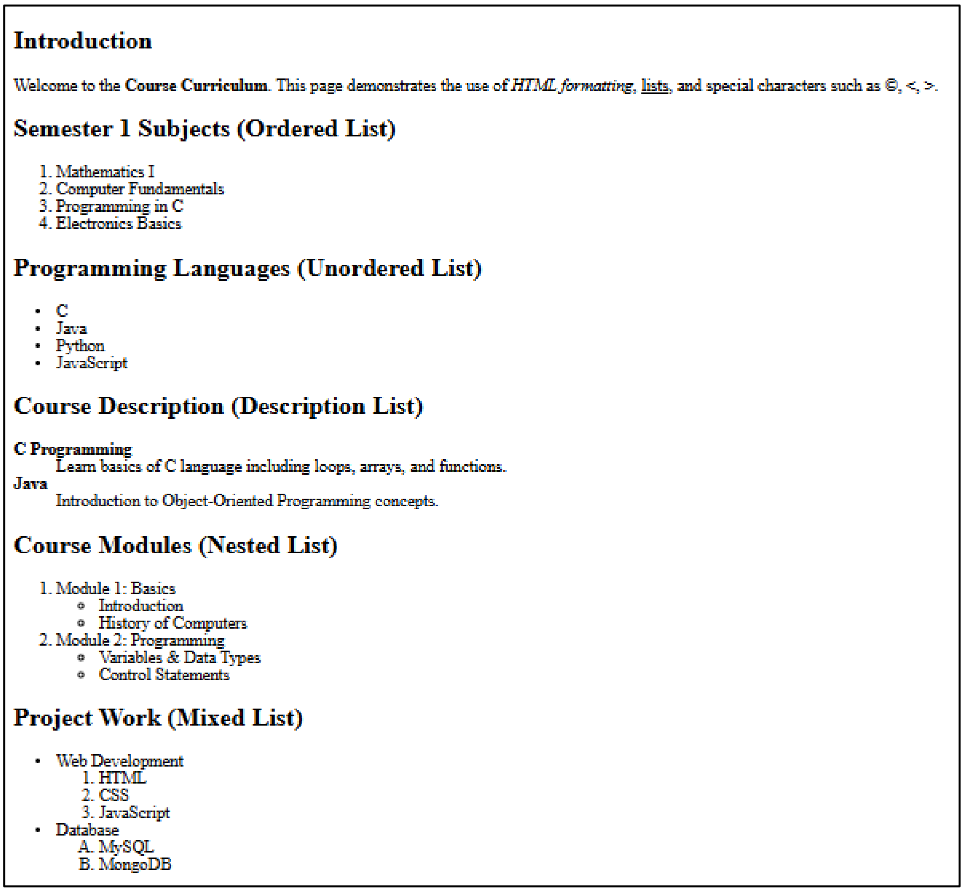
Theory

1. Headings (<h1>…<h6>): Used to display titles and subheadings in different sizes.
2. Paragraphs (<p>): Defines blocks of text content.
3. Formatting Tags:
    - <b> for bold text
    - <i> for *italic text*
    - <u> for <u>underlined text</u>
    - Special characters are displayed using HTML entities (e.g., ©, <, >).
4. Lists:
    - Ordered List (<ol>): Numbered items.
    - Unordered List (<ul>): Bulleted items.
    - Description List (<dl>): Term and description pairs.
    - Nested List: A list inside another list.
    - Mixed List: Combination of ordered, unordered, and nested lists.

Code Implementation

index.html

Output



Introduction

Welcome to the **Course Curriculum**. This page demonstrates the use of *HTML formatting*, lists, and special characters such as ©, <, >.

Semester 1 Subjects (Ordered List)

1. Mathematics I
2. Computer Fundamentals
3. Programming in C
4. Electronics Basics

Programming Languages (Unordered List)

- C
- Java
- Python
- JavaScript

Course Description (Description List)

C Programming
    Learn basics of C language including loops, arrays, and functions.
Java
    Introduction to Object-Oriented Programming concepts.

Course Modules (Nested List)

1. Module 1: Basics
    - Introduction
    - History of Computers
2. Module 2: Programming
    - Variables & Data Types
    - Control Statements

Project Work (Mixed List)

- Web Development
    1. HTML
    2. CSS
    3. JavaScript
- Database
    A. MySQL
    B. MongoDB

Outcome

- Successfully designed a basic course curriculum webpage.
- Implemented text formatting and different types of lists in HTML.
- Learned structuring and organizing content on a webpage.

Experiment 2

Create a Web Page for a News Website with Embedded YouTube Video, Hyperlinks, and Navigation Structure, and Insert Multimedia in a Webpage.

Objectives

1. To design a simple **news website homepage** using HTML and CSS.
2. To learn how to **embed YouTube videos** into a webpage.
3. To implement **hyperlinks and navigation menus** for easy browsing.
4. To insert **multimedia (images, videos, audio)** in a webpage.

Tools Required

- Computer / Laptop with any OS
- **Text Editor** (VS Code, Sublime Text, Notepad++)
- **Web Browser** (Chrome, Edge, Firefox)
- Internet connection (for YouTube embedding)

Theory

1. **Hyperlinks (<a> tag):**
   Hyperlinks allow navigation between pages or websites. Example:

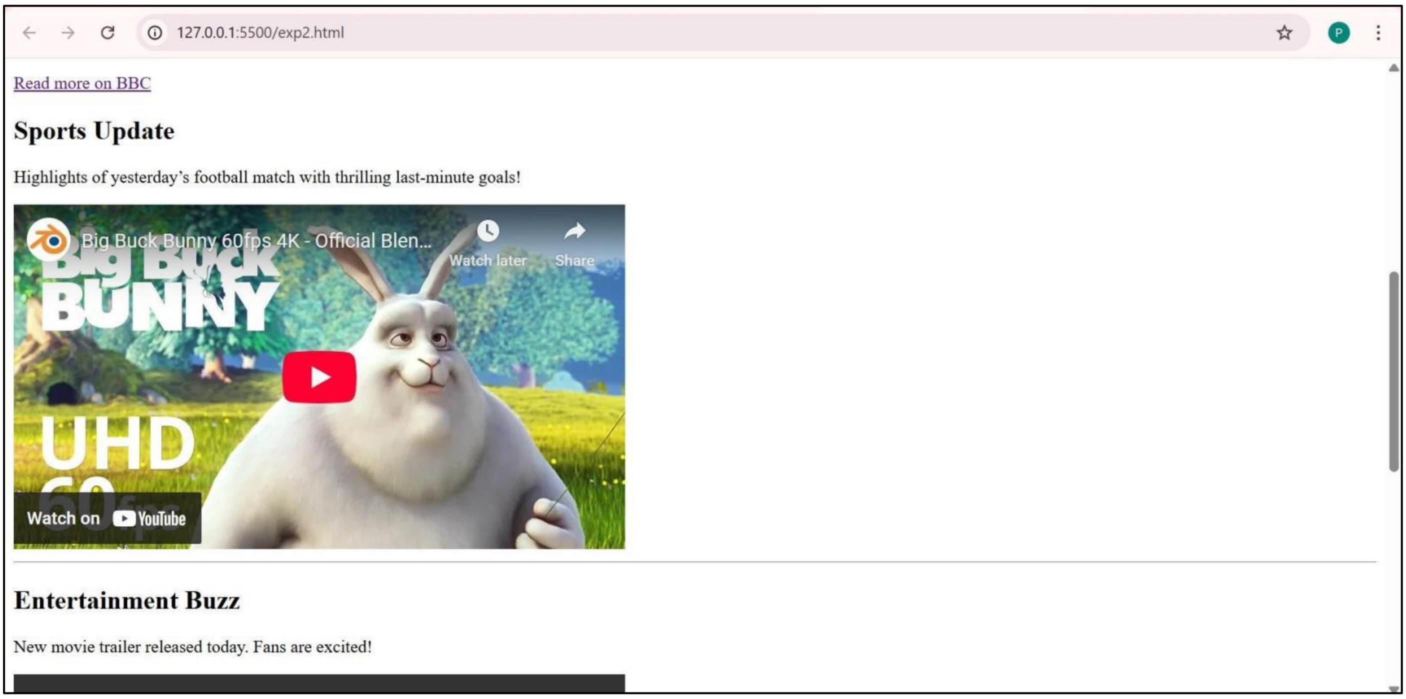
   ```
   <a href="https://www.bbc.com">Visit BBC News</a>
   ```
2. **Navigation Structure:**
   A <nav> element groups navigation links for better structure.
3. **Embedding YouTube Video:**
   YouTube provides an <iframe> code for embedding videos into a webpage.

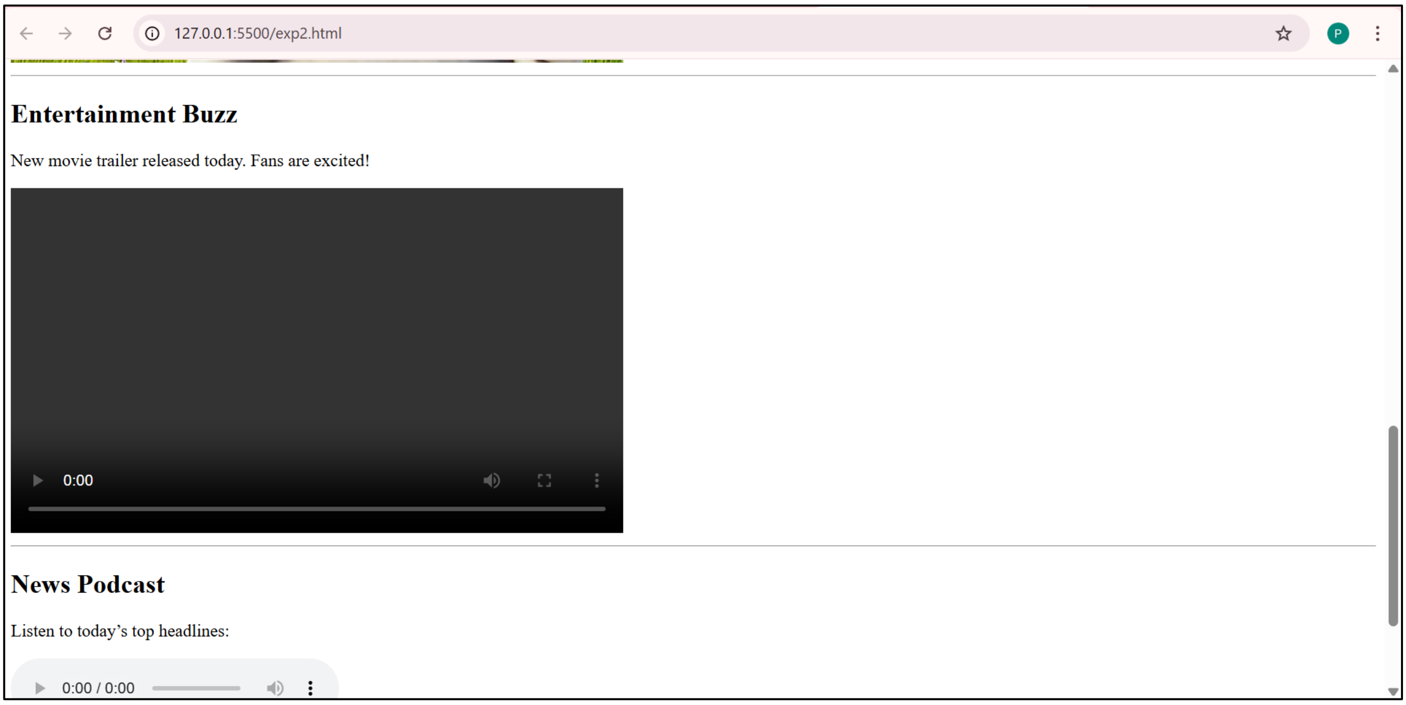
   ```
   <iframe src="https://www.youtube.com/embed/VIDEO_ID"></iframe>
   ```
4. **Multimedia in Webpages:**
   - **Image (<img>):** Display news images.
   - **Audio (<audio>):** Add podcasts or news audio clips.
   - **Video (<video>):** Insert local or hosted video files.

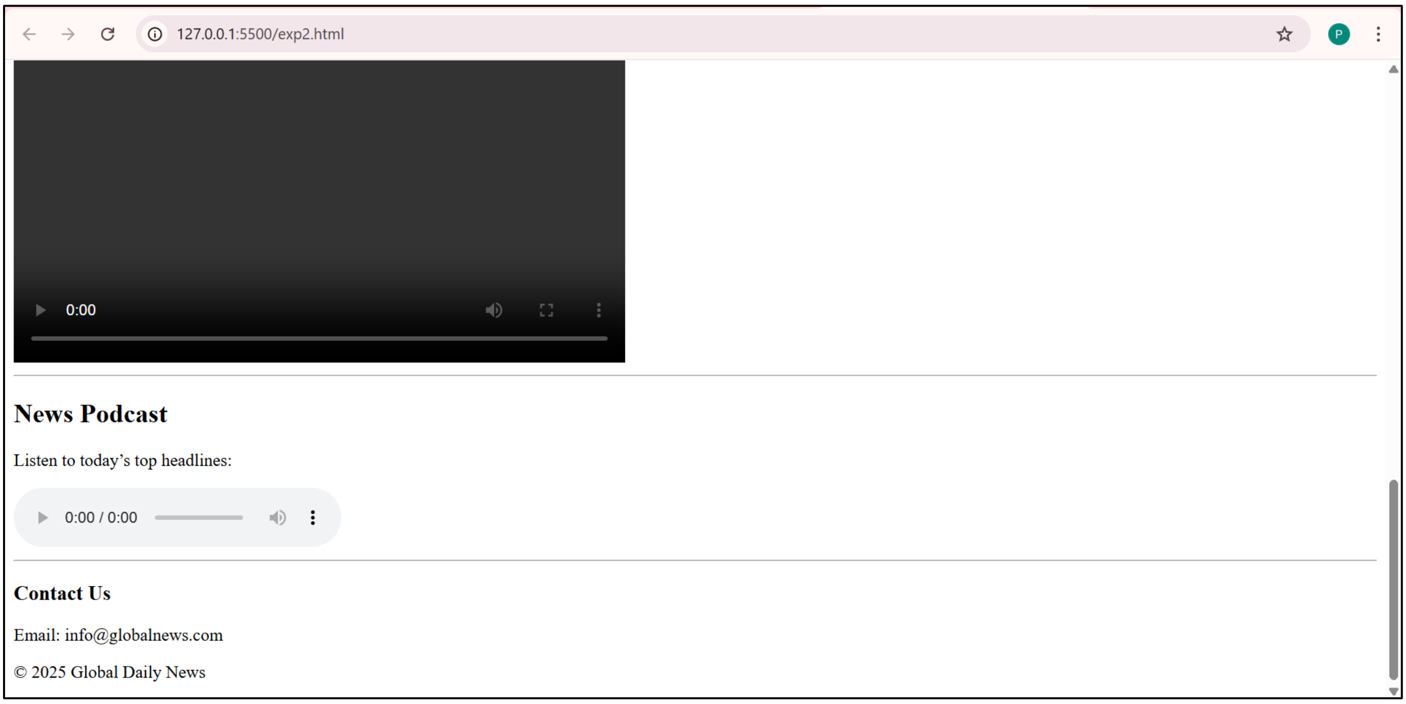
Code Implementation

index.html

Output


127.0.0.1:5500/exp2.html
Read more on BBC
Sports Update
Highlights of yesterday's football match with thrilling last-minute goals!
Big Buck Bunny 60fps 4K - Official Blen...
Watch later
Share
Watch on YouTube
Entertainment Buzz
New movie trailer released today. Fans are excited!


127.0.0.1:5500/exp2.html
Entertainment Buzz
New movie trailer released today. Fans are excited!
0:00
News Podcast
Listen to today's top headlines:
0:00 / 0:00



Outcome

- Successfully designed a **basic news website** layout.
- Implemented **hyperlinks, navigation, multimedia (image, audio, video), and embedded YouTube video**.
- Learned practical **webpage structuring techniques** for media-rich sites.

Experiment-3

Create a Web Page for College Timetable: Use of Tables with Rowspan, Colspan, Headers, Captions, and Nested Tables.

Objectives

1. To design a college timetable webpage using HTML tables.
2. To apply rowspan and colspan attributes for merging cells.
3. To use table headers (<th>), captions (<caption>), and formatting for readability.
4. To implement nested tables within a timetable cell.

Tools Required

- Computer / Laptop with any OS
- Text Editor (Notepad, VS Code, Sublime Text)
- Web Browser (Chrome, Edge, Firefox)

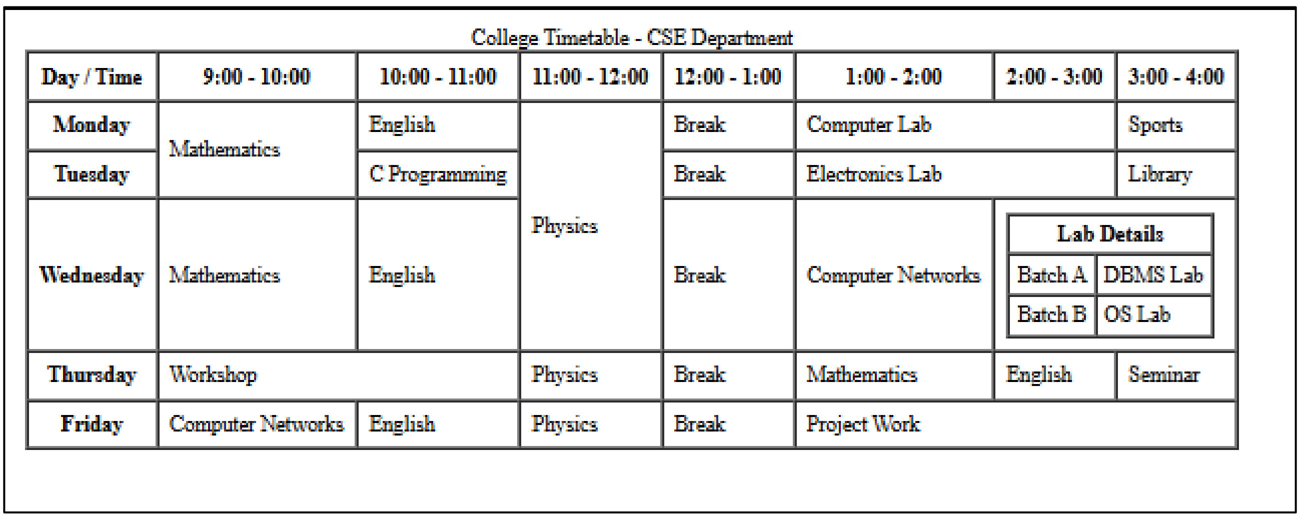
Theory

1. Table (<table>): Arranges data in rows and columns.
2. Rowspan: Merges a cell vertically across rows.
3. <td rowspan="2">Lab</td>
4. Colspan: Merges a cell horizontally across columns.
5. <td colspan="3">Break</td>
6. Header (<th>): Displays header cells in bold and centered.
7. Caption (<caption>): Defines the title of the table.
8. Nested Table: A table placed inside another table cell.

Code Implementation

index.html

Output:



<table>
<caption>College Timetable - CSE Department</caption>
<tr><th>Day / Time</th><th>9:00 - 10:00</th><th>10:00 - 11:00</th><th>11:00 - 12:00</th><th>12:00 - 1:00</th><th>1:00 - 2:00</th><th>2:00 - 3:00</th><th>3:00 - 4:00</th></tr>
<tr><th>Monday</th><td rowspan="2">Mathematics</td><td>English</td><td rowspan="3">Physics</td><td>Break</td><td colspan="2">Computer Lab</td><td>Sports</td></tr>
<tr><th>Tuesday</th><td>C Programming</td><td>Break</td><td colspan="2">Electronics Lab</td><td>Library</td></tr>
<tr><th>Wednesday</th><td>Mathematics</td><td>English</td><td>Break</td><td>Computer Networks</td><td colspan="2"><table><tr><th colspan="2">Lab Details</th></tr><tr><td>Batch A</td><td>DBMS Lab</td></tr><tr><td>Batch B</td><td>OS Lab</td></tr></table></td></tr>
<tr><th>Thursday</th><td colspan="2">Workshop</td><td>Physics</td><td>Break</td><td>Mathematics</td><td>English</td><td>Seminar</td></tr>
<tr><th>Friday</th><td>Computer Networks</td><td>English</td><td>Physics</td><td>Break</td><td colspan="3">Project Work</td></tr>
</table>

Outcome:

- Successfully created a college timetable webpage using HTML tables.
- Implemented rowspan, colspan, headers, captions, and nested tables.
- Learned to organize tabular data in a structured webpage format.

Experiment-4

Create Web page for Hotel room booking form: Use of input fields, labels, select, textarea, and buttons.

Objectives

1. To design a hotel room booking form using HTML.
2. To understand the use of input fields for collecting user details.
3. To apply labels for form accessibility.
4. To implement dropdown menus (select) and text areas.
5. To create functional buttons for form submission and reset.

Tools Required

- Computer / Laptop with any OS
- Text Editor (Notepad, VS Code, Sublime Text)
- Web Browser (Chrome, Edge, Firefox)

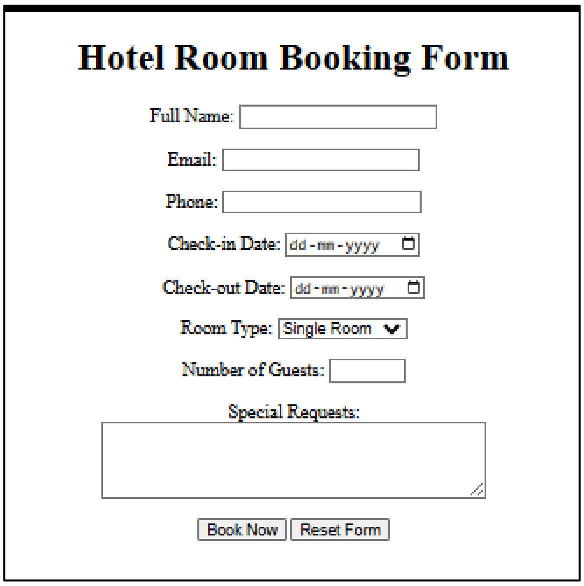
Theory

1. **Form (<form>)**: Defines a section for user input.
2. **Input Fields (<input>)**: Used for text, numbers, emails, dates, etc.
3. <input type="text" name="username">
4. **Labels (<label>)**: Connects text with form elements for better usability.
5. <label for="name">Name:</label>
6. **Select (<select>)**: Provides a dropdown menu.
7. <select><option>Single</option><option>Double</option></select>
8. **Textarea (<textarea>)**: Allows multi-line text input.
9. **Buttons (<button> or <input type="submit">)**: Used for form submission or reset.

Code Implementation

index.html

Output:



Outcome

- Successfully created a hotel room booking form webpage using HTML.
- Implemented input fields, labels, select, textarea, and buttons.
- Learned practical form design techniques for web applications.

Experiment 5

Create a Web Page for Student Registration Form with Email Validation by applying form attributes (method, action, placeholder, required).

Objectives

1. To design a **student registration form** using HTML.
2. To apply form attributes:
   - method → Defines how data is sent.
   - action → Defines where data is sent.
   - placeholder → Displays hint text.
   - required → Ensures mandatory fields are filled.
3. To implement **basic email validation** using the type="email" attribute.

Tools Required

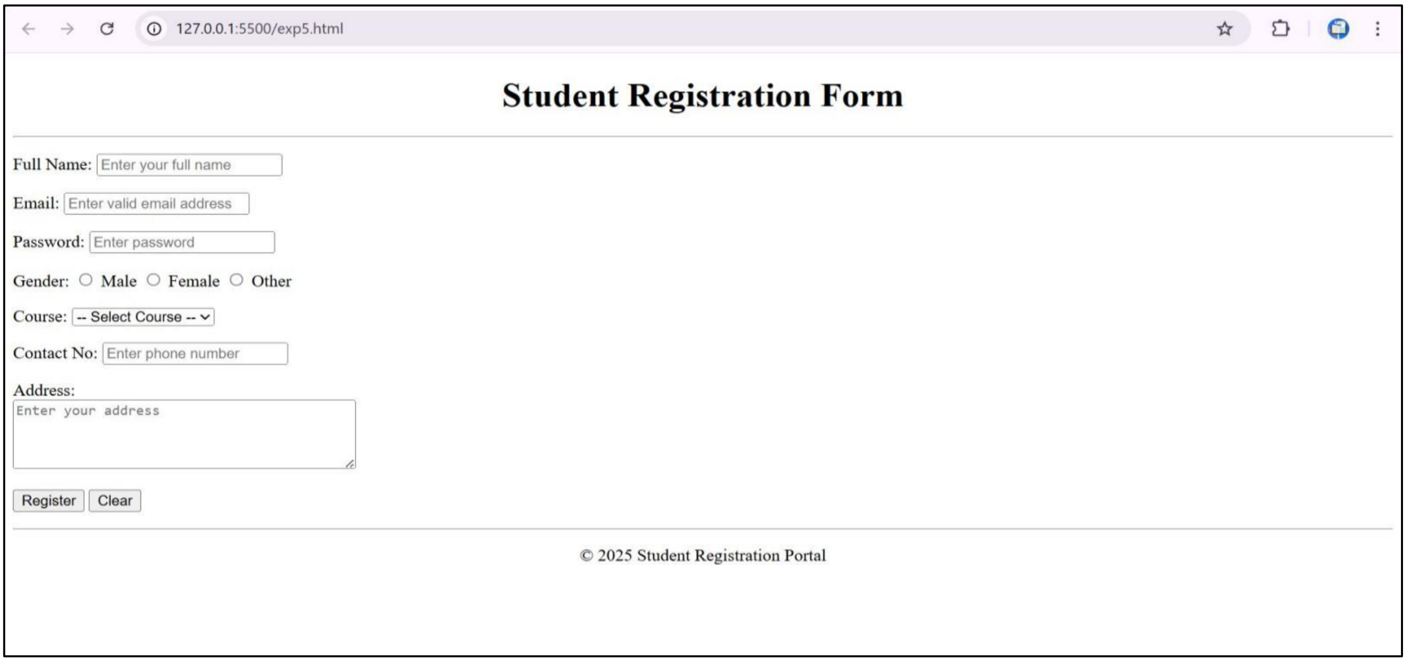
- Text Editor (Notepad, VS Code, Sublime Text)
- Web Browser (Chrome, Edge, Firefox)

Theory

- **HTML Forms** (<form>): Used to collect user input.
- **Form Attributes:**
    - method="post" → Sends form data securely.
    - action="submit.html" → Defines where the form data goes (dummy page here).
    - placeholder="Enter Name" → Shows hint text inside input box.
    - required → Makes the field compulsory.
- **Email Validation:**
    - type="email" → Ensures user enters data in email format (e.g., abc@example.com).

Code Implementation

Here's the HTML code:



Index.html**Output**

Outcome

- Successfully created a **Student Registration Form** using only HTML.
- Applied **form attributes**: method, action, placeholder, required.
- Implemented **email validation** using type="email".
- Learned to design a **structured form** without CSS.

Experiment 6

Create Web page for E-commerce Product Page: Use of CSS for style text, colors, fonts, margin, border, padding, width, and height.

Objectives

1. To design a simple **e-commerce product page** using HTML and CSS.
2. To demonstrate the use of **text, colors, and fonts** for better presentation.
3. To apply **margin, padding, and borders** for proper spacing and structure.
4. To set **width and height** for images and containers to create a neat layout.

Tools Required

- Computer / Laptop with any OS
- **Text Editor** (VS Code, Sublime Text, Notepad++)
- **Web Browser** (Chrome, Edge, Firefox)
- Internet connection (for YouTube embedding)

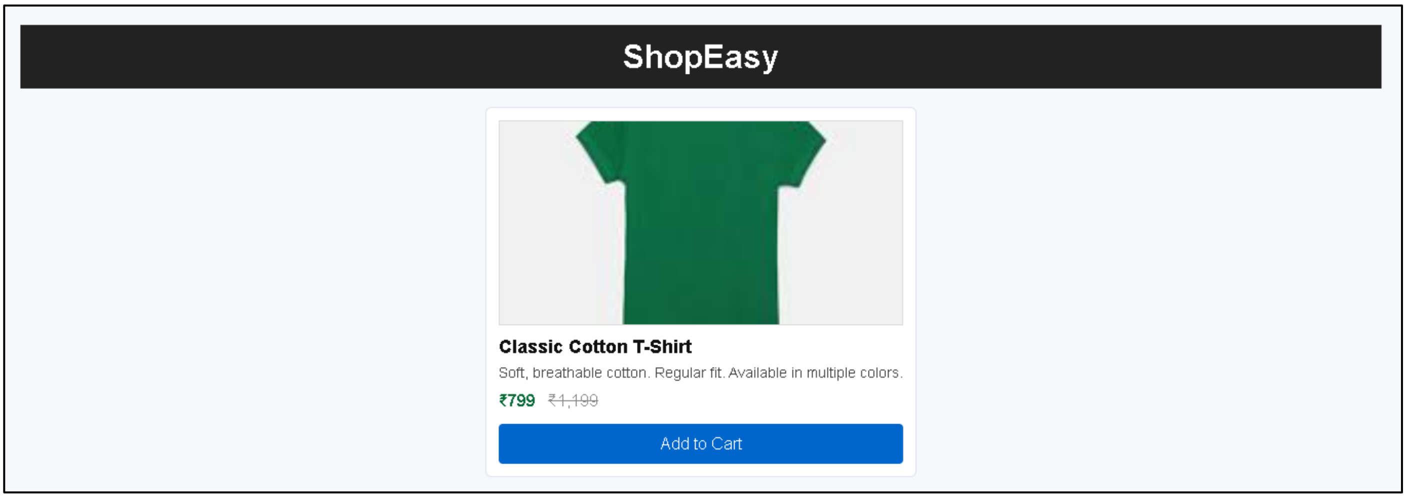
Code Implementation

Here's the **HTML code**:

index.html

```
}
```

Output:



Outcome

- Successfully created a **E-commerce Product Page** using only HTML & CSS
- Applied **CSS** for style text, colors, fonts, margin, border, padding, width, and height

Experiment 7

Create Web page for Navigation Menu: Use of block, inline, inline-block, float, and clear

---

Objectives

4. To design a simple navigation menu on a webpage.
5. To apply form attributes:

    - block for full-width sections like header
    - inline for text elements within a line.
    - inline-block for horizontal menu items that keep block features.

6. To apply float property for arranging navigation links side by side.
7. To use clear property to manage layout and prevent content overlap after floated elements.

Tools Required

- Text Editor (Notepad, VS Code, Sublime Text)
- Web Browser (Chrome, Edge, Firefox)

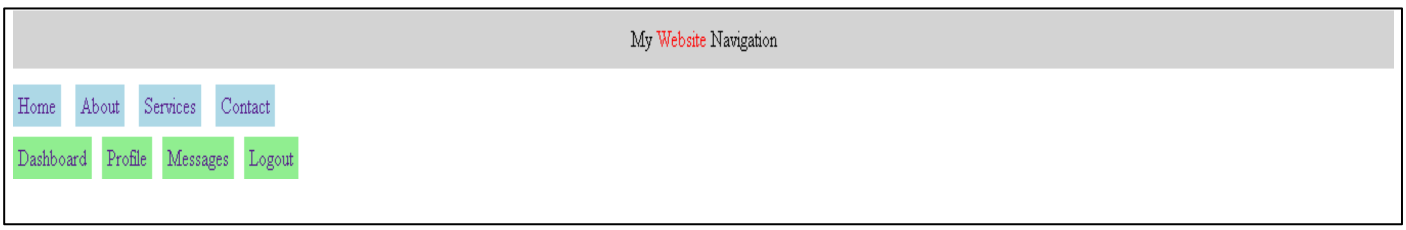
Code Implementation

Index.html

Here’s the **CSS code**:

style.css

Output



Experiment 8:

Create Web page for Student Profile Card: design layouts using flexbox (justify-content, align-items, flex-direction).

Objective: To design layouts using flexbox (justify-content, align-items, flex-direction).

Tools Required

- Text Editor (Notepad, VS Code, Sublime Text)
- Web Browser (Chrome, Edge, Firefox)

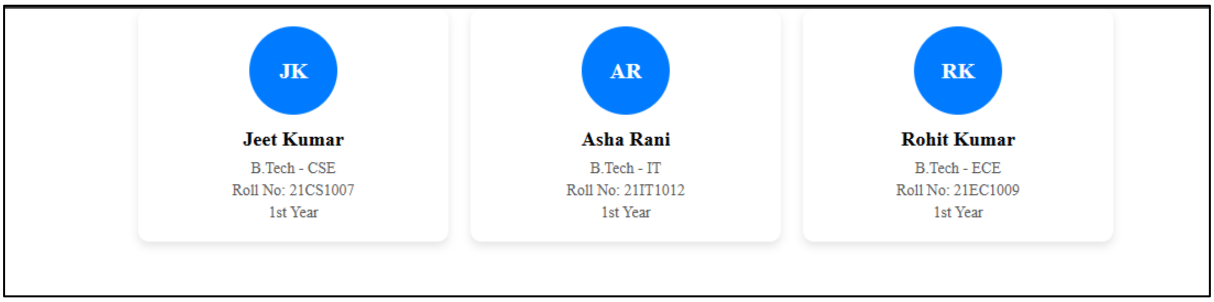
Theory:
Flexbox (**Flexible Box Layout**) is a CSS layout model that allows easy alignment and distribution of elements inside a container. It is especially useful for responsive design.
Key properties:

1. **display: flex;** → Defines a flex container.
2. **flex-direction:**
   - row → Items placed left to right (default).
   - column → Items placed top to bottom.
   - row-reverse / column-reverse → Reverse order.
3. **justify-content:** (aligns items on the main axis)
   - flex-start → Items at the beginning.
   - center → Items centered.
   - flex-end → Items at the end.
   - space-between, space-around, space-evenly → Distributes space.
4. **align-items:** (aligns items on the cross axis)
   - flex-start, center, flex-end, stretch, baseline.
5. **flex-wrap:** → Moves items to the next line if space is insufficient.

In this experiment, we apply **flexbox** to arrange multiple student profile cards neatly.

Code Implementation

Output:
Three student profile cards arranged in a row, aligned at the center of the page.



Outcome:

- By changing flex-direction, cards can be aligned in a **row** or **column**.
- Using justify-content and align-items, we can change the alignment of cards across the page.
- Successfully created a **Student Profile Card layout** using **Flexbox**.

Experiment 9:

Create Web page for Responsive Product Grid: create mobile-friendly designs with breakpoints.

Objective: To design a **Responsive Product Grid webpage** using **CSS Grid** and **media queries (breakpoints)** for mobile-friendly layouts.

Tools Required

- Text Editor (Notepad, VS Code, Sublime Text)
- Web Browser (Chrome, Edge, Firefox)

Theory:

A responsive web design adjusts automatically to different screen sizes (desktop, tablet, mobile).

- **CSS Grid:** Used to create structured layouts in rows and columns.
- **Breakpoints:** Specific screen width values defined in **media queries** to apply different CSS rules for different devices.

Common breakpoints:

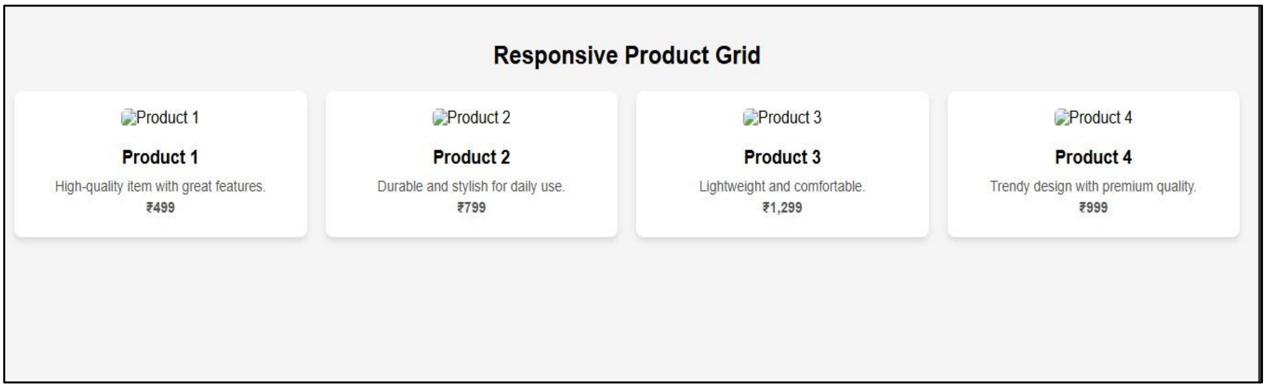
- **Desktop (large screens):** ≥1024px
- **Tablet (medium screens):** 600px – 1024px
- **Mobile (small screens):** ≤600px

In this experiment:

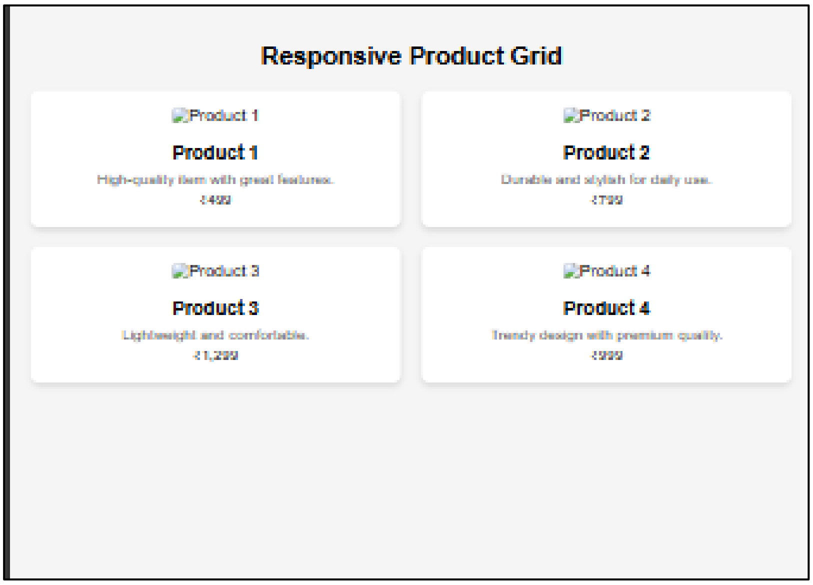
- Grid has **4 columns** on desktop,
- **2 columns** on tablets,
- **1 column** on mobiles.

Output:

- On **desktop:** 4 product cards per row.

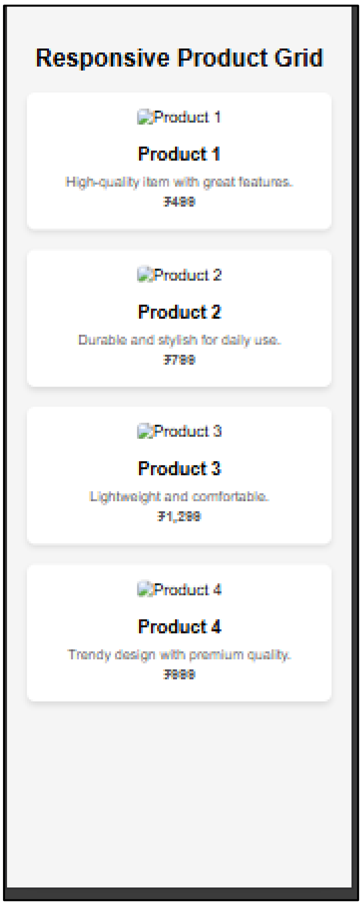


- On **tablet (≤1024px):** 2 product cards per row.



-

- On **mobile (≤600px):** 1 product card per row.



Outcome:

- The layout changes based on screen size. Breakpoints ensure readability and user-friendly design across devices.
- Successfully created a **Responsive Product Grid** webpage using **CSS Grid and media queries**.

Experiment 10

Create a Responsive E-commerce Style Navbar using HTML, CSS, Flexbox, and Media Queries.

Objectives

1. To design a **navigation bar (navbar)** for an e-commerce website.
2. To apply **CSS display properties** (block, inline, inline-block, float, clear).
3. To implement **Flexbox properties** (justify-content, align-items, flex-direction).
4. To use **relative units** (px, %, em, rem, vh, vw) for responsiveness.
5. To apply **media queries and breakpoints** for **mobile-first responsive design**.

Tools Required

- Text Editor (VS Code / Sublime Text / Notepad++)
- Web Browser (Chrome / Firefox / Edge)

Theory

- **Navigation Bar (Navbar):** A horizontal or vertical menu allowing users to browse site sections.
- **Display in CSS:**
  - block → element takes full width.
  - inline → element takes only needed width.
  - inline-block → inline element but allows width/height.
  - float and clear → for positioning elements.
- **Flexbox:**
  - display: flex; → creates a flex container.
  - justify-content → aligns items horizontally.
  - align-items → aligns items vertically.
  - flex-direction → row/column layout.
- **Responsive Design:**
  - **Units** (%, em, rem, vh, vw) adjust with screen size.
  - **Media Queries:** Example →
  - @media (max-width: 768px) { ... }

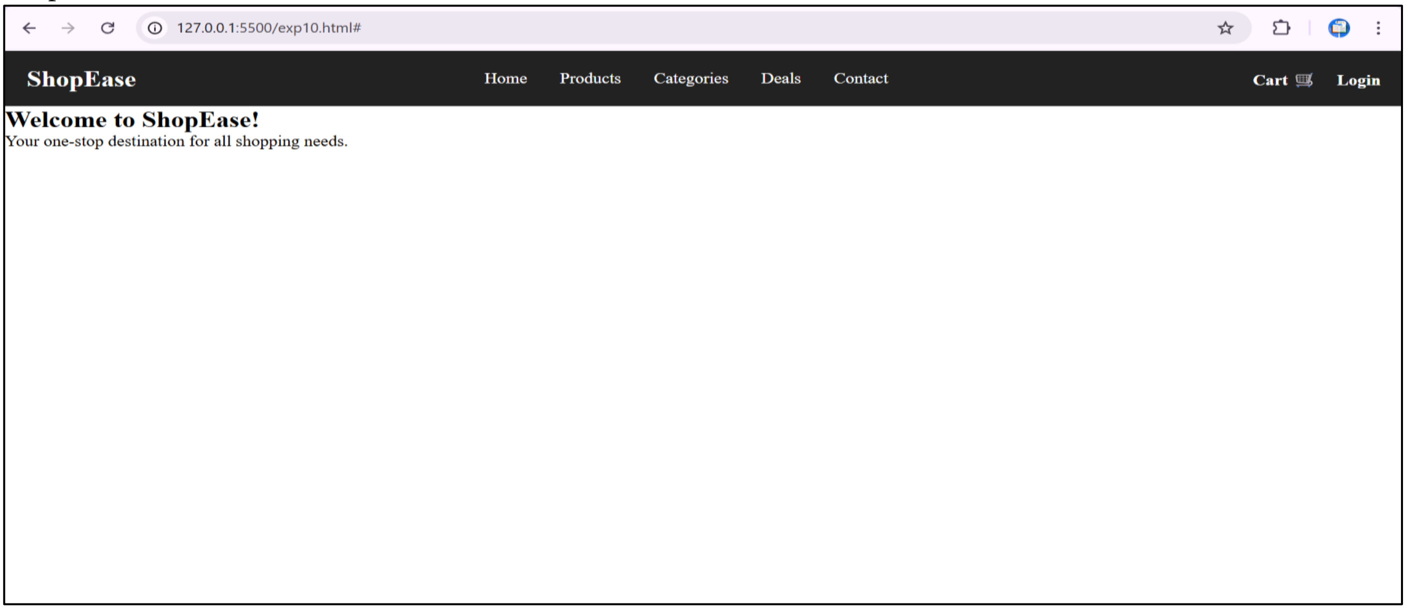
    lets us apply different styles on smaller devices.
- **Mobile-First Design:** Start with styles for small screens, then enhance for larger screens.

Code Implementation (HTML + CSS)

index.html

Here's the **CSS code**:
exp10style.css

Output



Outcome

A responsive **E-commerce style navbar** with logo, menu items, and cart/login buttons.

Experiment: Extended – Responsive E-commerce Navbar with Hamburger Menu

---

Objectives

- Create a responsive **e-commerce style navbar** with a hamburger menu.
- Learn how to toggle navigation menu on mobile screens using only **HTML + CSS**.
- Practice **flexbox**, **media queries**, and **mobile-first design**.

Tools Required

- Text Editor (VS Code / Notepad++ / Sublime).
- Web Browser (Chrome / Firefox / Edge).

Expected Outcome

- A responsive navbar with **logo + links + cart/login**.
- On **desktop/tablet** → full menu visible.
- On **mobile screen** → only logo + ☰ (hamburger). Clicking ☰ will show/hide the menu.

Theory

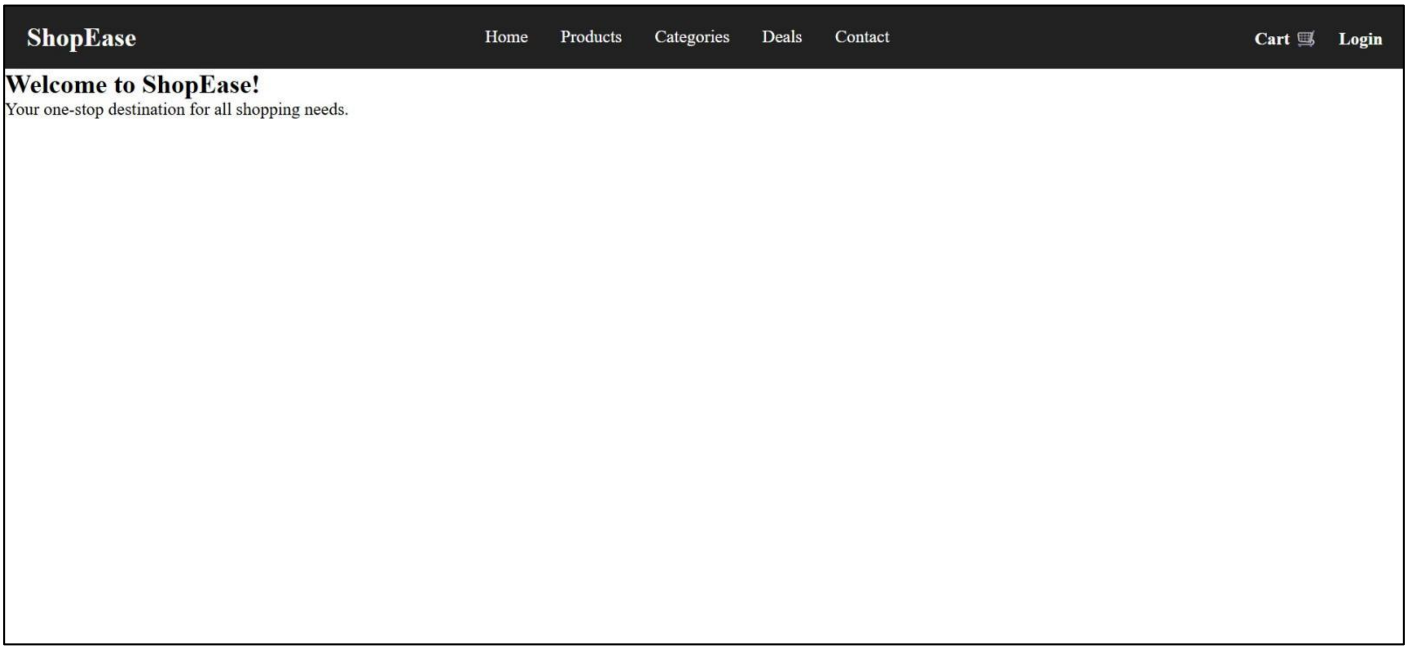
1. **Checkbox Hack**
   - A hidden <input type="checkbox"> is used.
   - A <label> with ☰ icon is linked to the checkbox.
   - When checkbox is checked → CSS: checked selector shows the menu.
2. **Flexbox Layout**
   - Navbar divided into **logo**, **links**, and **actions**.
   - justify-content: space-between; align-items: center; .
3. **Media Queries**
   - Hide links in small screens.
   - Show hamburger icon instead.
   - On click, menu slides down.

Code Implementation

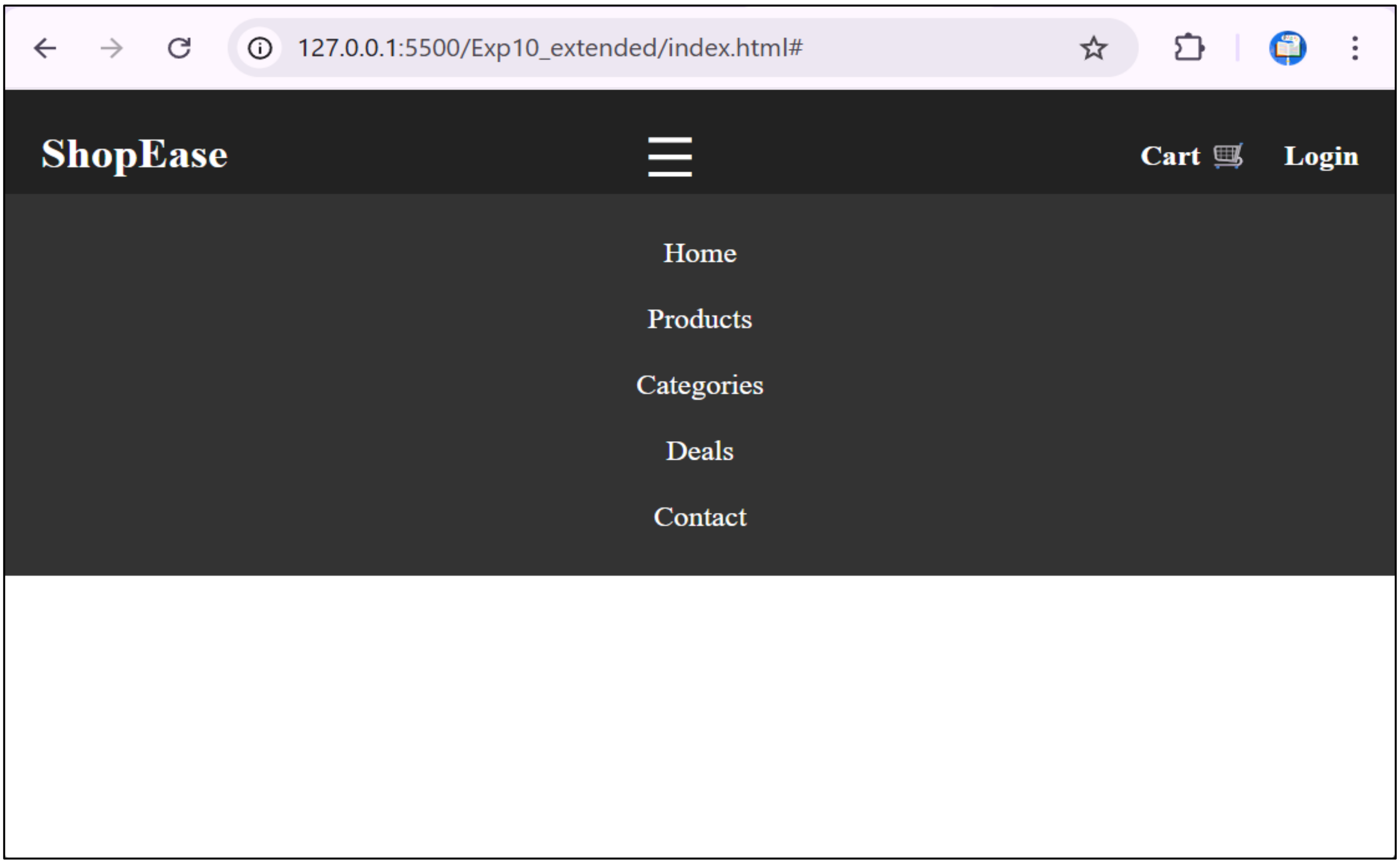
index.html

style.css

Output:



Mobile View (Collapsed)



Outcome

- Successfully created a **responsive e-commerce style navbar**.
- Used **display properties, flexbox, relative units, and media queries**.
- Implemented a **mobile-first design** with hamburger menu.
- Learned how to switch between **desktop and mobile layouts** using CSS.
